

2023

February 26, 2024 6:18 PM

• Question 1 [16 marks]

The output signal from an AM modulator is

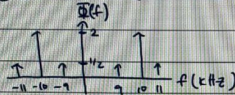
$$\varphi_{AM}(t) = \cos(2\pi \times 9,000t) + 4 \cos(2\pi \times 10,000t) + \cos(2\pi \times 11,000t).$$

- (1) Plot the spectrum of $\varphi_{AM}(t)$.
- (2) Write out a general expression for AM-LC signal. Determine the carrier signal $c(t)$, message signal $m(t)$, the corresponding modulation index for $\varphi_{AM}(t)$.
- (3) In $\varphi_{AM}(t)$, determine the carrier power P_c , the sideband power P_s , and power efficiency η .
- (4) Plot diagrams for a coherent (synchronous) and an Envelope Detector receiver structure. Specify the filter parameters.

(1) The output signal from AM modulator is

Carrier Power
E LE 846 m

$$1) \mathcal{F}(f) = \frac{1}{2} [\mathcal{S}(f-9000) + \mathcal{S}(f+9000)] + 2 [\mathcal{S}(f-10000) + \mathcal{S}(f+10000)] + \frac{1}{2} [\mathcal{S}(f-11000) + \mathcal{S}(f+11000)]$$



$$2) \varphi_{AM}(t) = (A_c + m(t)) \cos 2\pi f_c t$$

$$f_c = 10$$

$$\frac{A_c}{2} = 2 \Rightarrow A_c = 4 \quad \therefore c(t) = 4 \cos(2\pi \cdot 10 \times 10^3 t)$$

$$\frac{m(t)}{2} = \frac{1}{2} [\mathcal{S}(f-1) + \mathcal{S}(f+1)]$$

$$m(t) = 2 \cos(2\pi \cdot 10^3 t)$$

$$A_{max} = |4 + 2| = 6$$

$$A_{min} = |4 - 2| = 2$$

$$\mu = \frac{A_{max} - A_{min}}{2A_c}$$

$$\mu = \frac{6 - 2}{2(4)}$$

$$\mu = \frac{1}{2}$$

$$= \frac{1}{2}$$

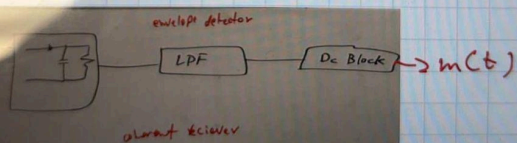
$$= 50\%$$

Also each side band has amplitude of $\frac{1}{2}$ so $\frac{1}{2} + \frac{1}{2} = 1$ but then to scale it back to $m(t)$ then multiply by 2. as $m=2$
 $\Rightarrow m(t) = 2 \cos(2\pi \cdot 1000t)$

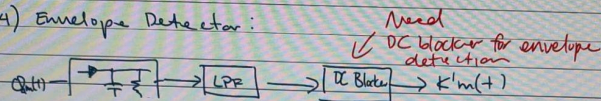
$$3) P_c = ?, P_s = ?, \eta = ?$$

$$P_c = (2)^2 \times 2 = 8, P_s = \left(\frac{1}{2}\right)^2 \times 2 + \left(\frac{1}{2}\right)^2 \times 2 = 1$$

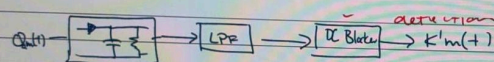
$$\eta = \frac{P_s}{P_s + P_c} = \frac{1}{1 + 8} = \frac{1}{9} = 11\%$$



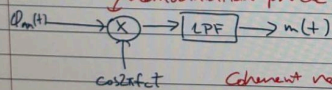
4) Envelope Detector:



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Envelope Detector, sends modulated signal through detector, LPF and DC blocker.



Coherent Receiver demodulates signal then sends through LPF

Both methods used to recover original signal

• Question 2 [20 marks]

A message signal $m(t)$ is given as

$$m(t) = 2 \cos(2\pi \times 100t) + \cos(2\pi \times 200t).$$

The above signal is modulated with a carrier: $c(t) = 2 \cos(2\pi \times 1,000t)$. Note that

$$\varphi_{SSB}(t) = m(t) \cos(\omega_c t) \pm m_h(t) \sin(\omega_c t)$$

- (1) Compute and sketch the magnitude spectrum and phase spectrum of $m(t)$, $M(f)$.
- (2) State and sketch transfer function of a Hilbert Transform $H(f)$ (amplitude response and phase response respectively).
- (3) What is Hilbert Transform $m_h(t)$ of $m(t)$? Plot magnitude spectrum and phase spectrum of $M_h(f)$.
- (4) Obtain and sketch the USB spectrum $\Phi_{USB}(f)$, and the corresponding waveform $\varphi_{USB}(t)$ by observing the spectrum.
- (5) Plot two methods to generate USB signal from a baseband signal. Specify the filter parameter with selective filter method, and phase-shift method.

$$M(f) = \delta(f+100) + \delta(f-100) + \frac{1}{2}[\delta(f+200) + \delta(f-200)]$$

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$$M(f) = \delta(f+100) + \delta(f-100) + \frac{1}{2}[\delta(f+200) + \delta(f-200)]$$

